DRAFT · COMPILED 17 OCT 2026 FROM PUBLIC SOURCES

From motor cortex to cursor: how Neuralink turns spikes into actions

Eight stages take the data from neurons firing in the hand-knob region to clicks, typing and a robot arm. Along the way it shrinks from about 205 Mbps to roughly 1 Mbps, and then to a few intents per second.

N1 implant · Neuralink App · PRIME / CONVOY trials

THE WHOLE PATH

Four stages run inside the skull and four run on the phone or laptop

INSIDE THE N1 IMPLANT

01
Neural source
Motor cortex fires when a hand movement is intended

02
Threads
1,024 electrodes on 64 polymer threads

03
Amplify + digitise
10-bit ADC at ~20 kHz per channel: **≈205 Mbps**

04
Spike detection
More than 200× reduction to spike events and features

EXTERNAL DEVICE

05
Radio + power
Through the scalp at **≈1 Mbps**, inductive charging

06
Decoding
Pretrained Mamba encoder with small decoder heads

07
Post-processing
Mode switcher maps intents to clicks, scroll, drag

08
Outputs
Cursor, 40 WPM typing, robot arm; speech and vision next

STAGES 1–2 · SIGNAL SOURCE AND SENSING

The implant listens to the part of the brain that plans hand movement

- Surgeons find the target before the operation with fMRI while the patient attempts hand movement. It sits on the precentral gyrus.
- Those neurons still fire when a movement is intended, even with a severed spinal cord.
- One implant in one hemisphere gives decodable signals for *both* hands.

- **N1:** 1,024 electrodes across 64 flexible threads, each thinner than a hair.
- The threads are too fine to place by hand, so the R1 robot inserts them. The 2019 robot inserted up to 6 threads a minute with 87.1% success.
- 2019 threads were polyimide with gold traces, 5–50 µm wide and 4–6 µm thick.
- **Next:** about 3,000 electrodes, plus design changes to keep threads in place.

STAGE 3 · ANALOG FRONT END AND ADC

Each channel is amplified, filtered and digitised on the implant

Custom low-power chips handle every electrode. The published numbers come from the 2019 research chip, which had 256 channels per chip:

PARAMETER	2019 ASIC
Gain	42.9–59.4 dB
Bandwidth	3 Hz – 27 kHz
Input noise	5.9 μVrms
ADC	10-bit @ 19.3 kHz
Power	5.2 μW / channel

Raw data the N1 produces

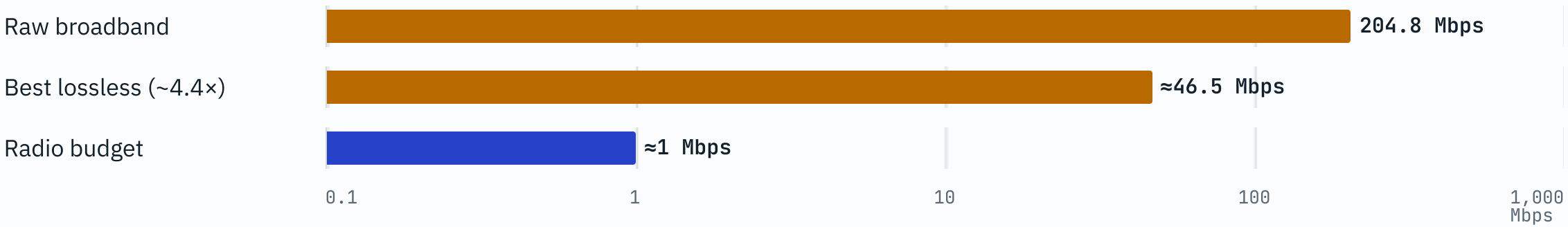
204.8 Mbps

1,024 ch × 20,000 samples/s × 10 bit = 204,800,000 bit/s

The production N1 chip specs aren't published. This rate comes from the Neuralink compression challenge figures.

STAGE 4 · ON-IMPLANT REDUCTION

The bottleneck: the radio carries about 1/200th of what the electrodes record



- The reduction has to run in real time, in under 1 ms, on less than 10 mW.
- Lossless compression stops around 4–5×, so the implant sends **spike events and features**, not waveforms.
- **Patent method:** filter the signal, fit a model, compare against thresholds, then package each spike with a timestamp.
- After threads pulled out in 2024, Neuralink retuned this step to pick up broader population activity.

STAGE 5 · WIRELESS LINK AND POWER

Spike data leaves the skull by radio, with no wires or connectors

- The processed data goes through the scalp to the Neuralink App on a phone or computer.
- The battery charges inductively from the N1 Charger. Nothing passes through the skin.
- The radio is usually described as Bluetooth Low Energy class. Neuralink hasn't confirmed this.

For comparison, the 2019 research rig

was wired: USB-C to a base station that streamed over 10 GbE. Wireless operation is what made at-home, all-day use possible. By Feb 2025 participants averaged 6.5 h a day.

STAGE 6 · DECODING, THEN AND NOW

Pretraining on 50,000 hours of brain data cut calibration time

BEFORE · PER-USER SUPERVISED

- Each user had a decoder for each feature, trained on calibration tasks.
- Neural signals drift, so decoders went stale.
- Users spent about 10 min a day recalibrating, or 55 min a week.

OCT 2026 · PRETRAINED ENCODER

- A self-supervised encoder trained on more than 50,000 h of unlabeled data.
- **30 s** of labeled data now matches what **3.5 min** of calibration used to give.
- Decoders last more than a week, and one still worked after 20 months.
- A model trained on one participant worked for another with 99% of its weights frozen.

STAGE 6 · INSIDE THE ENCODER

Spikes go in as tokens and intents come out of small heads

Spike tokens
Each spike is a token with a learned per-channel embedding (POYO-style)

→

Perceiver input
Compresses a variable number of spike tokens into a fixed set of latents

→

Mamba backbone
State-space model, fully causal and low-latency

→

Decoder heads
Cursor velocity, click probability, finger/key intent, 7+ arm DOF

TRAINING OBJECTIVE

Spatially masked auto-Poisson regression. The model sees half the channels and predicts spike counts on the other half. No labels are needed, so all of the participants' free-use hours become training data.

SCALE

Participant 1 alone contributed more than 9,000 hours, about 22.4 billion spikes. Neuralink is also testing decoding of target position directly, called "cursor teleportation", which exceeded 10 BPS in early tests.

STAGES 7–8 · POST-PROCESSING AND OUTPUTS

Decoded intents become ordinary input events

POST-PROCESSING

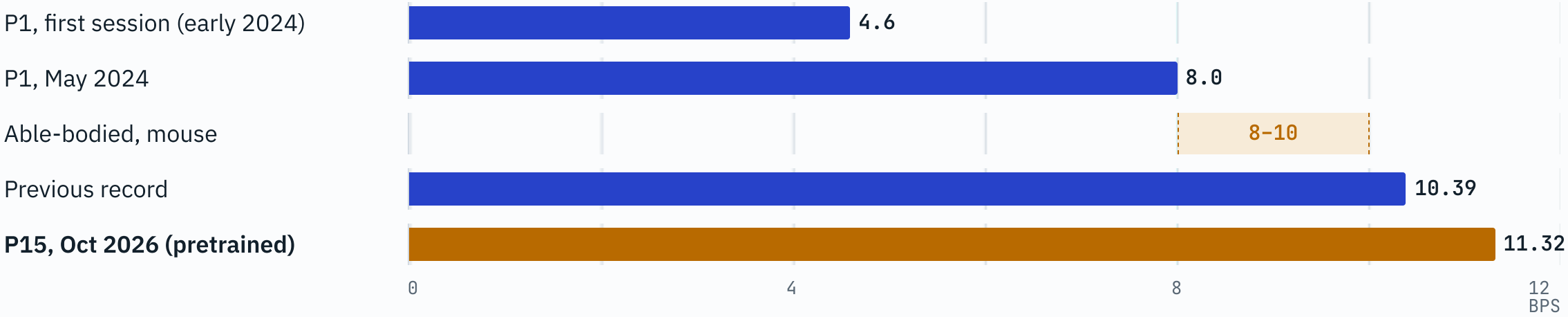
- Earlier versions used smoothing, low-velocity suppression and click filtering. Each step added lag.
- Pretrained decoders are stable enough that several of these steps were dropped.
- A mode switcher maps intents to left, right and middle click, scroll, zoom, pan and drag.

WHAT USERS DO WITH IT

- Control a computer or phone: chess, Civilization VI, Mario Kart, Fusion 360 CAD, Illustrator.
- Ten-finger "mind keyboard" typing at up to **40 WPM**.
- Robotic arm control in the CONVOY trial, for feeding and gesturing.
- **In trials or development:** speech at about 140 WPM (VOICE) and vision (Blindsight).

HOW THE OUTPUT IS MEASURED

Cursor control now beats a typical able-bodied mouse user



$$BPS = \log_2(N_{click} \cdot grid^2 - 1) \cdot \max(correct - incorrect, 0) / 60$$

As of Jan 2026, 21 participants were enrolled, with zero serious device-related adverse events reported. All figures are company-reported.

CONFIDENCE AND GAPS

Most of the path is documented, but stage 4 is inferred

STAGE	BASIS	CONFIDENCE
1, 2, 5, 6, 7, 8	Neuralink posts, 2019 peer-reviewed paper	primary
3 · front end	2019 research chip. Production N1 ASIC specs aren't published.	partial
4 · on-implant algorithm	Patent, compression-challenge numbers (page now offline), community analysis	secondary
5 · radio type	Press describes it as BLE-class. Not confirmed.	secondary
Performance	Company-reported, not independently checked	unverified

SOURCES

1. Musk E, Neuralink. An Integrated Brain-Machine Interface Platform With Thousands of Channels. *J Med Internet Res* 2019;21(10):e16194. doi:10.2196/16194
2. Neuralink. PRIME Study Progress Update (2024-04-12)
3. Neuralink. PRIME Update: User Experience (2024-05-08)
4. Neuralink. PRIME Update: Second Participant (2024-08-21)
5. Neuralink. A Year of Telepathy (2025-02-05)
6. Neuralink. Two Years of Telepathy (2026-01-28)
7. Neuralink. Pretraining on 50,000 Hours of Unlabeled Brain Data (2026-10-01)
8. Neuralink. Technology
9. Neuralink Compression Challenge README (offline)
10. US 2021/0012909 A1. Real-time neural spike detection
11. Spikey, community compression analysis